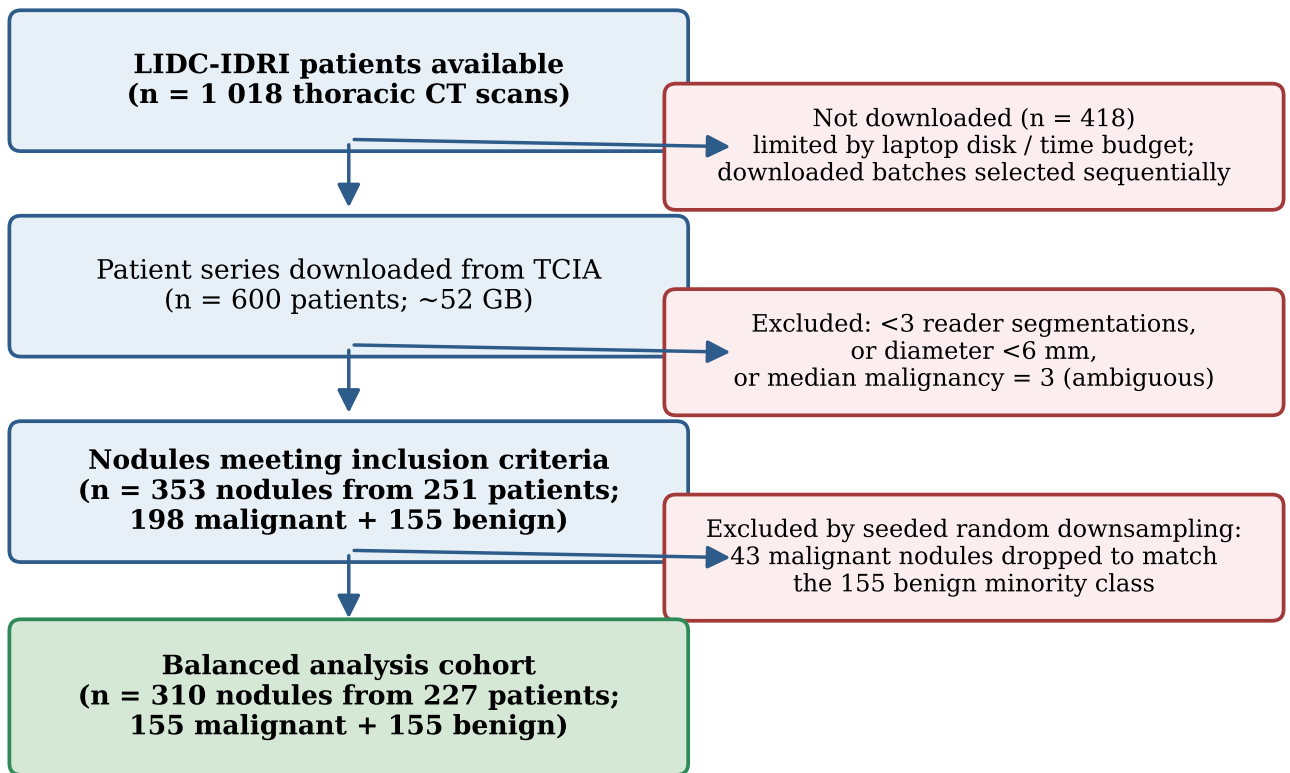


Supplementary Figure S1. Cohort flow.

Selection from LIDC-IDRI to the 310-nodule balanced analysis cohort.



Notes

1. The 6 mm median-diameter floor reflects the Fleischner Society 2017 actionable threshold for solid pulmonary nodules. The original pre-registered floor was 10 mm; the cascade to 6 mm followed the 29-benign yield at strict 10 mm and is documented in decisions.log #011. Sensitivity analysis at the original 10 mm floor is reported in Supplementary File S2.
2. The 353-nodule cohort (198 malignant + 155 benign from 251 patients) is retained for sensitivity analyses that test the effect of the balanced downsampling step (decisions.log #014).
3. Median pairwise inter-reader Dice across the 310-nodule cohort is 0.789 (mean 0.769, range 0.156-0.934), consistent with previously published LIDC inter-reader Dice values of 0.75-0.80 and supporting the external validity of the perturbation-magnitude calibration in Methods §2.3.
4. Per-nodule perturbation-condition count K varies between 7 and 12 across the cohort. Erosion retention rates by stratum are reported in Supplementary File S5; the full case-ID list is Supplementary File S4.